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Submission 261	
Title	SAT4RIDS: Self-Supervised Learning Adversarial Training Framework for Robust Network Intrusion Detection System
Paper	 (Nov 30, 01:34) (previous versions)
Author keywords	Network intrusion detection Self-supervised learning Adversarial training Packet header masking
Topics	Security and Privacy in Collaborative Systems
Abstract	Deep neural network (DNN) models are widely adopted in Network Intrusion Detection Systems (NIDS) but are vulnerable to adversarial attacks. Existing adversarial training methods primarily focus on supervised NIDS models and overlook semantic constraints of network traffic. To address these limitations, this paper proposes SAT4RIDS, a Self-supervised Adversarial Training Framework for Robust Intrusion Detection Systems. SAT4RIDS integrates adversarial training with self-supervised pre-training to learn robust traffic feature representations. We introduce a Packet Header Masking (PHM) method that ensures adversarial samples comply with protocol semantic constraints by applying adversarial perturbations only to the payload portion while preserving protocol-compliant packet headers. Experimental results on two intrusion detection datasets demonstrate that SAT4RIDS achieves state-of-the-art performance, with natural accuracy reaching 99.23%–99.86% and adversarial accuracy reaching 86.74%–87.53%, effectively defending against adversarial attacks while maintaining detection performance on normal traffic.
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